Infinite Dyck Paths Algebra for Unbounded Interval Temporal Logic

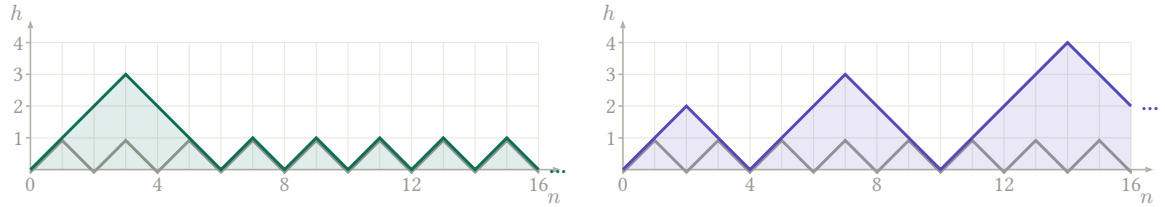
Introduction

> Instead of considering a finite total order, one can consider either an infinite (but discrete) total order or a more general partial order. In the former case, the combinatorial structure that is involved could be some notion of infinite Dyck path, that is, a Dyck path having infinite length. There are at least two different ways to formalize this suggestion: either working in the **set of all Dyck paths** (of any length), which is naturally partially ordered, and can be probably interpreted as some kind of infinitary limit object for the family of Dyck lattices; or introducing **Dyck paths having infinite support**, i.e., Dyck paths unbounded on the right which do not have a suffix consisting of the concatenation of an infinite number of peaks lying on the x-axis.
>
> — Ferrari

Therefore, we denote:

- **Model 1:** the set of all Dyck paths;
- **Model 2:** the set of Dyck paths with infinite support.

Both models live inside the same picture. Read an infinite word over $\Sigma = \{u, d\}^\omega$ as a lattice path and compare it to the bottom element $\bot = (ud)^\omega$, the infinite concatenation of peaks lying on the axis: the two models are exactly the two ways a path can behave against $\bot$ in the long run.



Ambient definitions

For a path $P = p_0 p_1 p_2 \ldots$, we define the function $h_P(n)$ that returns the height of P after its first n steps.

$$h_P(0) = 0, \qquad h_P(n) = |\{i < n \mid p_i = u\}| - |\{i < n \mid p_i = d\}|.$$

$$\mathcal{D}_\omega = \{P = (p_i)_{i \in \mathbb{N}} \in \{u, d\}^\omega \mid \forall n \in \mathbb{N}, h_P(n) \geq 0\}$$

Definition 1 (geometric order)**.** For $P, Q \in \mathcal{D}_\omega$ set

$$P \leq Q \quad \Leftrightarrow \quad \forall n \in \mathbb{N}, h_P(n) \leq h_Q(n),$$

and say that P lies *weakly below* Q. We call $\leq$ the *geometric order*, since it compares the two drawings pointwise. It restricts to each D_n, and $(D_n, \leq)$ is the Dyck lattice of order n of [1].

We import the vocabulary of [1] about specific Dyck paths, transposed to infinite words. A *return* of a path P is an index $n > 0$ with P crossing the x axis i.e. $h_P(n) = 0$; a *peak* is a pair of consecutive steps $p_n p_{n+1} = ud$, and a *hill* is a peak at height 0, i.e. one with $h_P(n) = 0$. A *pyramid* is a block $u^k d^k$ ($k \geq 1$) starting and ending on the axis, so a hill is the pyramid with $k = 1$. Finally, a *factor* of P is a block of consecutive steps running between two consecutive returns (or from the origin to the first

return); a hill is also called a *trivial factor*. In the finite case every Dyck path is the juxtaposition of its factors; here the same holds for the prefix of P up to its last return, and a path with infinitely many returns is the juxtaposition of infinitely many factors. The atom π_i are the path whose factors are all hills except for a single pyramid $uudd$.

Definition 2 (limit path)**.** In this language, we denote $\bot$ as the unique path all of whose factors are hills.

$$\bot = (ud)^\omega, \qquad h_\bot(n) = \begin{cases} 0 \text{ if } & n = 0 \bmod 2 \\ 1 \text{ if } & n = 1 \bmod 2 \end{cases}.$$

The *support* of a path measures how far it strays from the floor $\bot$: it collects the times at which P does not sit exactly where $\bot$ does. Since $\bot$ is nothing but hills, this amounts to recording where P does something other than a hill: a path of finite support is one whose factors are hills from some point on — a finite Dyck path followed by an infinite run of hills — whereas a path of infinite support keeps producing nontrivial factors arbitrarily late. This is precisely the line separating Model 1 from Model 2.

$$\mathrm{supp}(P) = \{n \in \mathbb{N} \mid h_P(n) \neq h_\bot(n)\}, \qquad \mathrm{supp}_{\mathrm{at}}(P) = \{i \geq 1 \mid h_P(2i) > 0\}.$$

The second variant samples only the even times, and there $\bot$ is on the axis: since paths never go below 0, "differing from $\bot$" and "being strictly above the axis" say the same thing. The next two lemmas show that this sampling loses nothing, so we may use whichever variant is convenient.

Lemma 3. $\mathrm{supp}_{\mathrm{at}}(P) = \{i \mid 2i \in \mathrm{supp}(P)\}$.

Lemma 4. $|\mathrm{supp}(P)| < \infty \Leftrightarrow |\mathrm{supp}_{\mathrm{at}}(P)| < \infty$.

Lemma 5. $i \in \mathrm{supp}_{\mathrm{at}}(P) \Leftrightarrow \pi_i \leq P$, *where* $\pi_i = (ud)^{i-1}uudd(ud)^\omega$.

Ferrari represents a path of D_n by an antichain of intervals of $[n-1]$, namely the orders of the atoms lying below each join-irreducible in its decomposition [1]. We keep the intervals, but not the antichain.

Definition 6 (interval representation)**.** Let Int denote the set of finite nonempty intervals $I = [a, b]$ of $\mathbb{N}_{\geq 1}$, ordered by inclusion. For $I = [a, b] \in \mathrm{Int}$ put

$$J_I = (ud)^{a-1}u^{b-a+2}d^{b-a+2}(ud)^\omega,$$

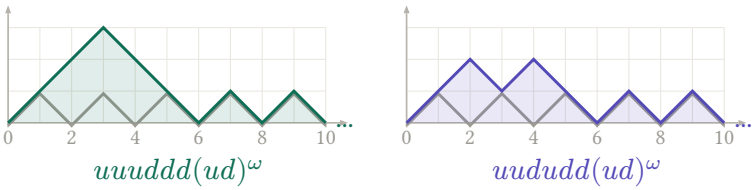
the path all of whose factors are hills except for a single pyramid, and set

$$\hat{F}_P = \{I \in \mathrm{Int} \mid J_I \leq P\}.$$

Each J_I is join-irreducible in $\mathcal{D}_\omega$, and $J_{[i,i]} = \pi_i$; hence $\mathrm{supp}_{\mathrm{at}}(P) = \{i \mid [i, i] \in \hat{F}_P\}$: the support records exactly the singletons of $\hat{F}_P$, and forgets every wider interval.

Lemma 7. $J_I \leq J_{I'}$ *if and only if* $I \subseteq I'$. *In particular* $\hat{F}_P$ *is a down-set of* Int.

Example. Knowing $\mathrm{supp}_{\mathrm{at}}(P)$ is not enough to know P. Both paths below have $\mathrm{supp}_{\mathrm{at}} = \{1, 2\}$, yet $uududd(ud)^\omega < uuuddd(ud)^\omega$.



$uuuddd(ud)^\omega$ $\qquad$ $uududd(ud)^\omega$

The order is exactly what the picture shows: at every abscissa the right path sits weakly below the left one, and that pointwise comparison of heights is all that $\leq$ means.

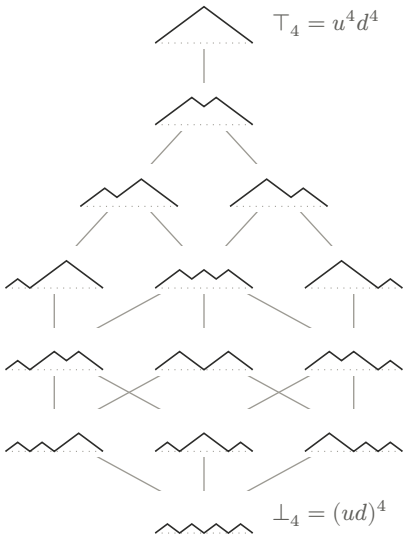


Figure 3: The Dyck lattice D_4, drawn in the geometric order.

The two are nevertheless different paths. The left one is a single pyramid, hence join-irreducible with atom-interval $[1, 2]$, while the right one is $\pi_1 \vee \pi_2$; what separates them is the down-set, $\hat{F} = \{[1, 1], [2, 2], [1, 2]\}$ against $\hat{F} = \{[1, 1], [2, 2]\}$, of which $\mathrm{supp}_{\mathrm{at}}(P)$ retains only the singletons. So $\mathrm{supp}_{\mathrm{at}}(P)$ tells us whether P has finite support, but not which path P is; only $\hat{F}_P$ does that.

Lemma 8. *Pointwise* min *and* sup *of height functions of paths in $\mathcal{D}_\omega$ are height functions of paths in $\mathcal{D}_\omega$; hence $(\mathcal{D}_\omega, \leq)$ is a complete lattice with $\bot = (ud)^\omega$, $\top = u^\omega$, and binary meets distribute over arbitrary joins, so $\mathcal{D}_\omega$ is a complete Heyting algebra.*

Dyck paths lattice

Each $(D_n, \leq)$ is a finite distributive lattice. The Figure 3 illustrate the lattice for $n = 4$. The bottom is the path of four hills $\bot_4$, the top the single pyramid $\top_4$, and an edge joins two paths when one covers the other in the geometric order.

These finite lattices are the stages of a single tower. To pass from one stage to the next we leave the path alone and append a hill at the right end by the inclusion function ι defined below. The Figure 4 draws the case ι_3.

$$\iota_n : D_n \to D_{n+1}, \qquad \iota_n(P) = Pud.$$

Appending a hill is also the extension that fixes the floor, sending $\bot_n = (ud)^n$ to $\bot_{n+1}$, which keeps the whole tower compatible with the reference path against which support is measured. In the interval representation the map does nothing at all: the intervals representing P are unchanged and only the ambient index set grows from $[n-1]$ to $[n]$, the new atom π_n failing to lie below Pud because Pud has returned to the axis at abscissa $2n$. Appending hills lets a path of one stage be compared with a path of any later one, and two paths should count as equal when the tower eventually makes them so.

Definition 9 (eventual equality)**.** For $n \leq k$ let $\iota_{n,k} : D_n \to D_k$ denote the composite $\iota_{k-1} \circ \cdots \circ \iota_n$, with $\iota_{n,n} = \mathrm{id}$; that is, $\iota_{n,k}(P) = P(ud)^{k-n}$. On the disjoint union

$$\coprod_{n \in \mathbb{N}} D_n = \{(n, P) \mid n \in \mathbb{N}, P \in D_n\}$$

define

$$(n, P) \sim (m, Q) \quad \Leftrightarrow \quad \exists k \geq \max(n, m) : \iota_{n,k}(P) = \iota_{m,k}(Q).$$

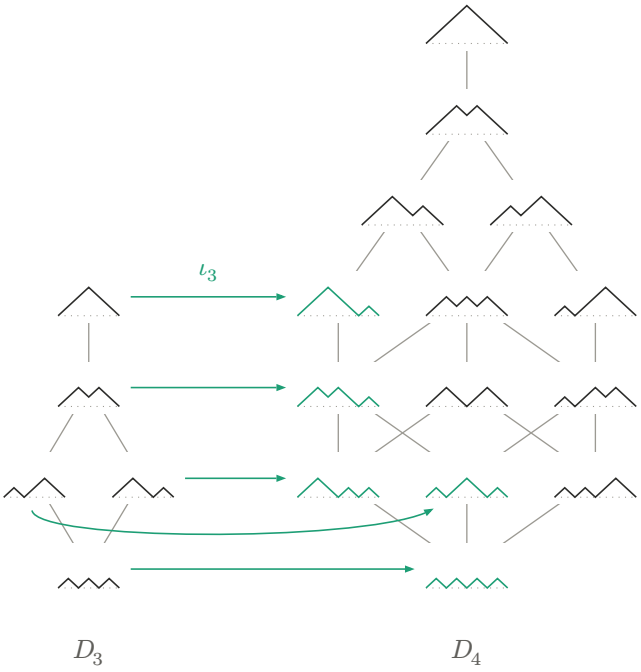


Figure 4: The case $\iota_3 : D_3 \hookrightarrow D_4$.

The relation $\sim$ is obviously an equivalence relation. Its classes are what let paths of different lengths be compared at all: $P \in D_n$ and $Q \in D_m$ with $n \neq m$.

intro D_ω lattice and figure Figure 5

Model 1

We restate that Model 1 corresponds to the set of all Dyck paths (of any length), which is naturally partially ordered, and can probably be interpreted as some kind of infinitary limit object for the family of Dyck lattices [2], [3], [4].

Lemma 10. *ι_n is injective.*

Lemma 11. *ι_n is an order-embedding preserving function.*

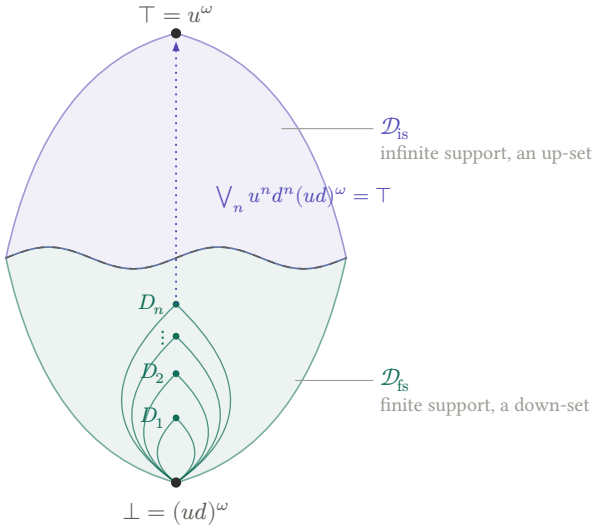


Figure 5: The two strata of $\mathcal{D}_\omega$, and the tower $D_1 \subset D_2 \subset \cdots D_\omega$.

The infinitary limit object $\mathcal{D}_{\text{colim}}$ for the family of Dyck lattices, denoted $\operatorname{colim} D_n$, is the colimit of the chain $D_0 \xrightarrow[\iota]{} D_1 \xrightarrow[\iota]{} D_2 \to \cdots$ along the maps ι_n: the smallest object in which every stage sits compatibly. Therefore, the $\mathcal{D}_{\text{colim}}$ is simply the increasing union of the tower quotiented by $\sim$.

Definition 12 (Direct limit of D_n)**.** Writing $[(n, P)]$ for the $\sim$-class of (n, P), put

$$\mathcal{D}_{\text{colim}} = \left(\coprod_{n \in \mathbb{N}} D_n \right) / \sim, \qquad c_n : D_n \to \mathcal{D}_{\text{colim}}, \quad c_n(P) = [(n, P)],$$

and equip $\mathcal{D}_{\text{colim}}$ with

$$[(n, P)] \leq [(m, Q)] \quad \Leftrightarrow \quad \exists k \geq \max(n, m) : \iota_{n,k}(P) \underset{D_k}{\leq} \iota_{m,k}(Q),$$

$$[(n, P)] \vee [(m, Q)] = \left[\left(k, \iota_{n,k}(P) \vee_{D_k} \iota_{m,k}(Q) \right) \right],$$

and

$$[(n, P)] \wedge [(m, Q)] = \left[\left(k, \iota_{n,k}(P) \wedge_{D_k} \iota_{m,k}(Q) \right) \right]$$

for any $k \geq \max(n, m)$, and $\bot = [(0, \varepsilon)]$ with ε the empty path.

An element of $\mathcal{D}_{\text{colim}}$ is thus a finite Dyck path together with the convention that the flat ground to its right does not count. Letting that ground run forever is exactly a subset of $\mathcal{D}_\omega$.

$$j : \mathcal{D}_{\text{colim}} \hookrightarrow \mathcal{D}_\omega \qquad j([(n, P)]) = P(ud)^\omega.$$

Proposition 13. *j is an order-embedding, and $j(\mathcal{D}_{\text{colim}}) = \{P \in \mathcal{D}_\omega \mid |\text{supp}(P)| < \infty\}$.*

Algebraic representation

Model 1 has a second possible description, it is an initial algebra. Recall that an algebra for an endofunctor F is just a map $F(X) \to X$ and a coalgebra is a map $X \to F(X)$; neither carries the associativity that an algebra over a monad, or a coalgebra over a comonad, would supply. Least and greatest fixpoints of F are written μF and νF respectively.

Definition 14 (initial algebra)**.** An *F-algebra* is a pair (X, α) with $\alpha : F(X) \to X$, and a morphism of F-algebras $(X, \alpha) \to (Y, \beta)$ is a map $f : X \to Y$ with $f \circ \alpha = \beta \circ F(f)$. An algebra $(\mu F, \text{in})$ is *initial* when for every F-algebra (X, α) there is exactly one such f, that is a unique

$$\text{fold}(\alpha) : \mu F \to X \quad \text{with} \quad \text{fold}(\alpha) \circ \text{in} = \alpha \circ F(\text{fold}(\alpha)).$$

cite

Recall finite Dyck words are constructed with the grammar $S ::= \varepsilon \mid uSdS$ generates, and that grammar is a signature functor.

Definition 15 (Dyck initial algebra)**.** Let $F : \text{Set} \to \text{Set}$ be

$$F(X) = 1 + X \times X,$$

with constructors ε (i.e. $\emptyset$) for the terminal object 1 and $u(-)d(-)$ for the product $X \times X$, the latter reading a pair (A, B) as the word $uAdB$.

Proposition 16. *F is finitary, so its initial algebra exists and is computed by the initial chain $\emptyset \to F(\emptyset) \to F^2(\emptyset) \to \cdots$, which converges at ω:*

$$\mu F = \bigcup_{k \in \mathbb{N}} F^k(\emptyset),$$

n	terms of μF	word	count
0	ε	ε	$C_0 = 1$
1	$u\,\varepsilon\,d\,\varepsilon$	ud	$C_1 = 1$
2	$u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon)$	$udud$	$C_2 = 2$
	$u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon$	$uudd$	
3	$u\,\varepsilon\,d\,(u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon))$	$ududud$	$C_3 = 5$
	$u\,\varepsilon\,d\,(u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon)$	$uduudd$	
	$u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,(u\,\varepsilon\,d\,\varepsilon)$	$uuddud$	
	$u\,(u\,\varepsilon\,d\,(u\,\varepsilon\,d\,\varepsilon))\,d\,\varepsilon$	$uududd$	
	$u\,(u\,(u\,\varepsilon\,d\,\varepsilon)\,d\,\varepsilon)\,d\,\varepsilon$	$uuuddd$	

Table 1: The correspondance of μF and Catalan number $C_n = \frac{1}{n+1}\binom{2n}{n}$

with $\mathrm{in} : F(\mu F) \to \mu F$ *the pair of constructors. Its elements are the finite Dyck words before any quotient.*

Let $\sigma : F(\mathbb{N}) \to \mathbb{N}$ be the F-algebra $\sigma(*) = 0$ and $\sigma(a, b) = a + b + 1$, and set

$$\ell := \mathrm{fold}(\sigma) : \mu F \to \mathbb{N}, \quad \ell(\varepsilon) = 0, \quad \ell(uAdB) = \ell(A) + \ell(B) + 1,$$

the unique map counting the uses of the constructor $u(-)d(-)$, i.e. the semilength.

Proposition 17. *The fibres of ℓ are the strata of the tower: $\ell^{-1}(n)$ is exactly the carrier of D_n, so that*

$$\mu F = \coprod_{n \in \mathbb{N}} \ell^{-1}(n) \cong \coprod_{n \in \mathbb{N}} D_n,$$

and $\left|\ell^{-1}(n)\right| = C_n$:

$$1, 1, 2, 5, 14, 42, 132, \ldots$$

the Catalan numbers.

The maps ι_n glue into a single unary operation on μF. Concatenation is not a constructor of μF, so appending a hill is not available as such and has to be written as a structural recursion:

$$r(\varepsilon) = u\,\varepsilon\,d\,\varepsilon, \qquad r(uAdB) = uAd\,r(B),$$

which on words is $A \mapsto A(ud)$ and restricts to ι_n on each fibre $\ell^{-1}(n)$.

Lemma 18. *r is injective and $\ell(r(A)) = \ell(A) + 1$. Hence eventual equality (Definition 9 (eventual equality)) takes, on μF alone, the form*

$$A \sim B \quad \Leftrightarrow \quad B = r^{\ell(B) - \ell(A)}(A) \quad (\ell(A) \le \ell(B)),$$

with no pair (n, P) and no cofinal stage left.

Comparing semilengths in $r^i(A) = r^j(B)$ forces $i - j = \ell(B) - \ell(A)$, and cancelling r^j by injectivity leaves a single application of r on one side: the number of hills to append is never searched for, it is the length difference.

Example. In $uAdB$ the subterm A sits inside the first factor and r never touches it; only B, the rest of the word, is walked. So r descends the right spine of the term, which is exactly the list of factors of the word, and appends one hill to that list, in one step per factor:

$$[] \mapsto [ud], \qquad [uudd] \mapsto [uudd, ud], \qquad [ud, ud] \mapsto [ud, ud, ud], \qquad [uudd, ud] \mapsto [uudd, ud, ud].$$

Thus $uudd \sim uudd\,udud$ ($k = 2$), whereas $udud \nsim uudd$: equal semilengths force $k = 0$, and the two terms differ.

Proposition 19. *$\mathrm{im}(r)$ is the set of Dyck words whose last factor is a hill, every $A \in \mu F$ is uniquely $A_0(ud)^k$ with $A_0 \notin \mathrm{im}(r)$, and the classes of $\sim$ are the r-orbits. Quotienting the initial algebra by that relation therefore recovers the colimit of the inclusion chain of Dyck lattices, and does so with a canonical section:*

$$\mathcal{D}_{\mathrm{colim}} = \mu F/\sim \cong \mu F \setminus \mathrm{im}(r),$$

the hill-reduced finite Dyck paths, with $j(A_0) = A_0(ud)^\omega$.

Model 2

$$\mathcal{D}_{\infty\text{-supp}} = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}, \qquad \mathcal{D}_\omega = j(\mathcal{D}_{\mathrm{colim}}) \sqcup \mathcal{D}_{\infty\text{-supp}}.$$

Proposition 20. *$\mathcal{D}_{\infty\text{-supp}}$ is an up-set of $\mathcal{D}_\omega$, closed under arbitrary joins, but closed under neither binary meets (interleaved combs: P, Q with pyramids at odd resp. even slots have $P \wedge Q = \bot$) nor pseudonegation; dually, $j(\mathcal{D}_{\mathrm{colim}})$ is closed under $\vee, \wedge, \bot$ but not under $\sim$, $\nrightarrow$, or directed sups ($\bigvee_n u^n d^n (ud)^\omega = u^\omega$).*

$$R(P) = \{i \geq 1 \mid h_P(2i) = 0\}, \qquad \mathcal{D}_{\infty\text{-supp}} = \underbrace{\{|R(P)| = \infty\}}_{\text{recurrent}} \sqcup \underbrace{\{|R(P)| < \infty\}}_{\text{transient}}.$$

Unbounded Interval Temporal Logics

$$\mathcal{D}_{\mathrm{fs}} := \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| < \infty\}, \qquad \mathcal{D}_{\mathrm{is}} := \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}.$$

$$\mathcal{D}_\omega = \mathcal{D}_{\mathrm{fs}} \sqcup \mathcal{D}_{\mathrm{is}}.$$

Logic of Model 1

$$\mathcal{D}_1 := \operatorname*{colim}_n D_n \cong \mathcal{D}_{\mathrm{fs}} = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| < \infty\}.$$

$$\bot = (ud)^\omega \in \mathcal{D}_1, \qquad \top \notin \mathcal{D}_1.$$

$$P, Q \in \mathcal{D}_1 \quad \Rightarrow \quad P \wedge Q \in \mathcal{D}_1, \qquad P \vee Q \in \mathcal{D}_1.$$

$$h_{P \wedge Q}(n) = \min\big(h_P(n), h_Q(n)\big), \qquad h_{P \vee Q}(n) = \max\big(h_P(n), h_Q(n)\big).$$

$$\mathrm{supp}(P \wedge Q) = \mathrm{supp}(P) \cap \mathrm{supp}(Q), \qquad \mathrm{supp}(P \vee Q) = \mathrm{supp}(P) \cup \mathrm{supp}(Q).$$

$$\mathcal{D}_1 = \text{distributive lattice with bottom, without top.}$$

$$\neg\exists\top \quad \Rightarrow \quad \neg\exists \text{ global Heyting implication.}$$

$$\mathcal{L}_1 = \{\bot, \wedge, \vee\}.$$

Logic of Model 2

$$\mathcal{D}_2 := \mathcal{D}_{\mathrm{is}} = \{P \in \mathcal{D}_\omega \mid |\mathrm{supp}(P)| = \infty\}.$$

$$(ud)^\omega \notin \mathcal{D}_2, \qquad \top = u^\omega \in \mathcal{D}_2.$$

$$P, Q \in \mathcal{D}_2 \quad \Rightarrow \quad P \vee Q \in \mathcal{D}_2.$$

$$\mathrm{supp}(P \vee Q) = \mathrm{supp}(P) \cup \mathrm{supp}(Q), \qquad |\mathrm{supp}(P)| = \infty \quad \Rightarrow \quad |\mathrm{supp}(P \vee Q)| = \infty.$$

$$P, Q \in \mathcal{D}_2 \quad \not\Rightarrow \quad P \wedge Q \in \mathcal{D}_2.$$

$$\mathrm{supp}(P \wedge Q) = \mathrm{supp}(P) \cap \mathrm{supp}(Q), \qquad |\mathrm{supp}(P)| = |\mathrm{supp}(Q)| = \infty \quad \not\Rightarrow \quad |\mathrm{supp}(P) \cap \mathrm{supp}(Q)| = \infty.$$

$$\bot \notin \mathcal{D}_2, \qquad \wedge \text{ is not closed.}$$

$$\mathcal{D}_2 = \text{join-semilattice with top}, \qquad \mathcal{L}_2 = \{\top, \vee\}.$$

$$\neg\exists \bot, \qquad \neg\exists \text{ global } \wedge, \qquad \neg\exists \text{ global } \to, \qquad \neg\exists \text{ global negation.}$$

Completion

$$\mathcal{D}_\omega = \mathcal{D}_{\mathrm{fs}} \sqcup \mathcal{D}_{\mathrm{is}}, \qquad \mathcal{D}_1 \to \mathcal{D}_\omega \leftarrow \mathcal{D}_2.$$

$$\mathcal{D}_1 \cong \mathcal{D}_{\mathrm{fs}} \subseteq \mathcal{D}_\omega, \qquad \mathcal{D}_2 = \mathcal{D}_{\mathrm{is}} \subseteq \mathcal{D}_\omega.$$

$$\bot = (ud)^\omega, \qquad \top = u^\omega.$$

$$h_{\sup_{i\in I} P_i}(n) = \sup_{i\in I} h_{P_i}(n), \qquad h_{\inf_{i\in I} P_i}(n) = \inf_{i\in I} h_{P_i}(n).$$

$$P \wedge \left(\sup_{i\in I} Q_i\right) = \sup_{i\in I}(P \wedge Q_i).$$

$$\mathcal{D}_\omega = \text{frame} = \text{complete Heyting algebra.}$$

$$P \to Q = \sup\{R \in \mathcal{D}_\omega \mid P \wedge R \leq Q\}, \qquad \neg P = P \to \bot.$$

$$\mathcal{L}_\omega = \{\bot, \top, \wedge, \vee, \to, \neg\} + \{\sup_{i\in I}, \inf_{i\in I}\}.$$

$$\mathrm{Idl}(\mathcal{D}_1) \cong \mathcal{D}_\omega.$$

Final overview

Bibliography

[1] L. S. Ferrari, "Dyck Algebras, Interval Temporal Logic, and Posets of Intervals," *ArXiv*, 2015, [Online]. Available: https://api.semanticscholar.org/CorpusID:14279208

[2] L. Ferrari and R. Pinzani, "Lattices of lattice paths." [Online]. Available: https://arxiv.org/abs/math/0411427

[3] L. Ferrari and E. Munarini, "Lattices of Paths: Representation Theory and Valuations." [Online]. Available: https://arxiv.org/abs/0905.3764

[4] J.-L. Baril, S. Kirgizov, and M. Naima, "A lattice on Dyck paths close to the Tamari lattice." [Online]. Available: https://arxiv.org/abs/2309.00426